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OptiGob: A decision support tool for Agriculture, Forestry and Other Land Use transitions

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Software

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Summary

`OptiGob` is a Python-based tool designed to explore configurations of Ireland's agriculture, forestry, and other land use (AFOLU) sectors out to 2100. The main purpose is to assist users in the assessment environmental and economic impact pathways, based on different land use transition pathways under varying assumptions about agricultural, afforestation, emissions abatement, and carbon dioxide removal (CDR) strategies. OptiGOB combines data outputs from the GOBLIN Lite for agriculture and land use, from FERS-CBM for forestry, and from the LCAD2.0 for anaerobic digestion, to generate scenarios that respect biophysical constraints. `OptiGob` provides a flexible, customizable, tool for researchers, policymakers, and educators to explore environmental and economic trade-offs associated with land use transition pathways.

Statement of need

Ireland's agricultural landscape is dominated by grassland, supporting extensive dairy and suckler beef production. Investment in Ireland's bioeconomy requires substantial changes to Agriculture, Forestry and Other Land Use (AFOLU) sector. `OptiGob` allows users to manipulate critical levers within the AFOLU sector that determine transition pathways and systematically explore their impacts. `OptiGob` utilises data generated by `GOBLIN Lite` (Duffy et al., 2022, 2024), FERS-CBM (Black et al., 2025; Kurz et al., 2008), and LCAD2.0 (Martinez-Acre, 2025) to calculate outputs from agricultural production, forestry, and anaerobic digestion, respectively.

`OptiGob` model estimates the net greenhouse gas (GHG) emissions from land use change, based on user input, and estimates available emissions and land budget for grass-based livestock (Dairy and Suckler cow) production. Pyomo (Bynum et al., 2021; Hart et al., 2011) is used to optimise livestock populations, while respecting area commitment (afforestation, anaerobic digestion, BECCS, protein crops) and emission (CO_2e in the case of Net-Zero, and CH_4, along with Net-Zero N_2O and CO_2 in the case of split gas targets) constraints.

The GOBLIN modelling framework has been used in recent studies on pathways to net-zero for the AFOLU sector (Bishop et al., 2024; Duffy et al., 2022; Henn et al., 2025). The `OptiGob` tool gives users the opportunity to gain significant insight into the potential cross-cutting synergies and trade-offs with substantially reduced overhead and complexity.

Model Overview

Figure 1 illustrates the architecture of the `OptiGob` model. User-defined parameters are provided via JSON or CSV input files and parsed by a data manager, which supplies values to the relevant

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sub-modules. The central `OptiGob` class orchestrates the overall model flow, coordinating sub-modules for emissions, land area, and economic outcomes. These in turn call modules representing forestry, livestock, bioenergy, other land uses, and static agricultural components such as crops, sheep, pigs, poultry, and protein crops, as well as substitution effects.

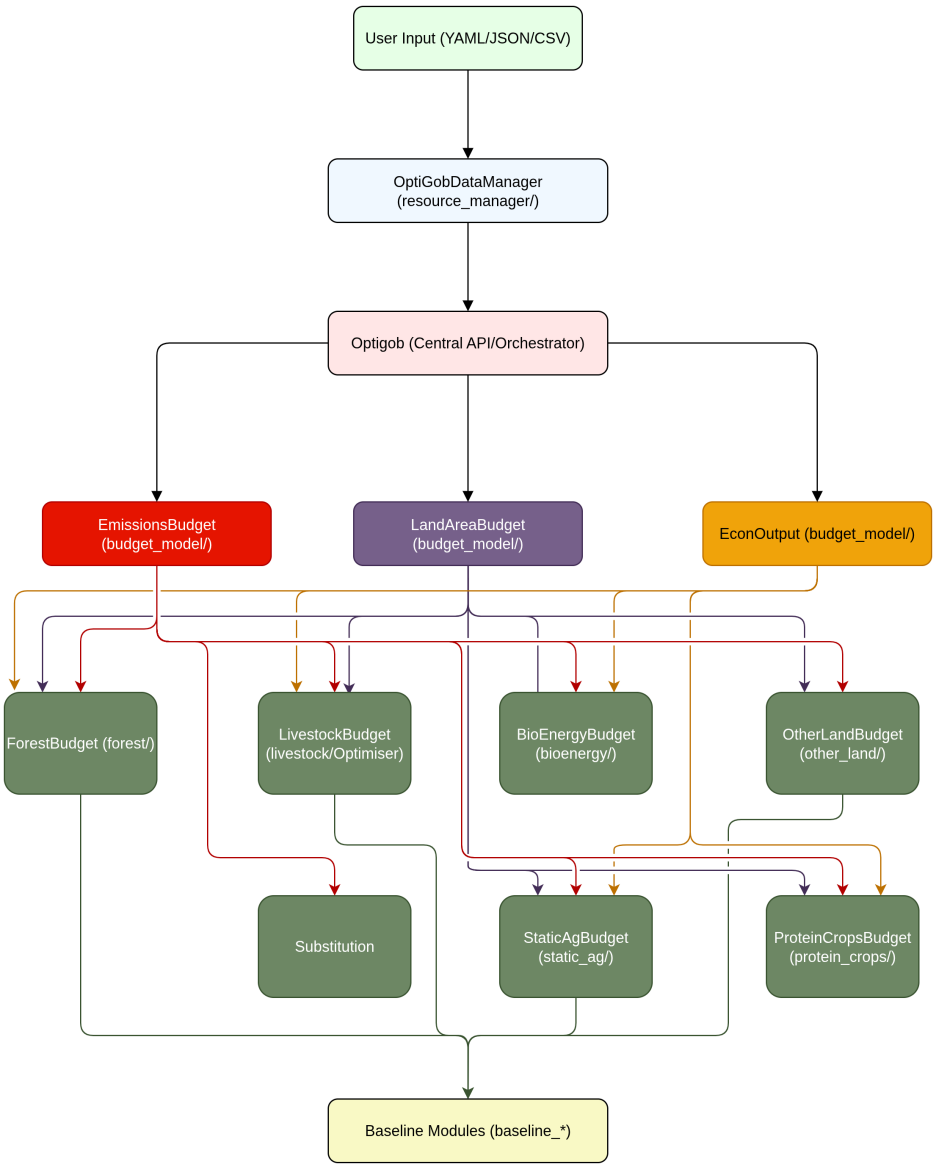

Figure 1: OptiGob Architecture.

`OptiGob` enables exploration of Ireland's AFOLU sector under alternative assumptions about productivity, emissions abatement, land use change, and carbon dioxide removal (CDR). Key user inputs include: the dairy-to-beef cow ratio, level of agricultural abatement, agricultural productivity growth, afforestation rate and harvest intensity, wetland and organic soil restoration, biomethane deployment, land allocation to protein crops and willow bioenergy, and inclusion of BECCS (Bioenergy with Carbon Capture and Storage).

The model workflow begins by estimating potential CDR from forestry, harvested wood

products, and BECCS to 2050. Fixed emissions from land use, crop production, pigs, poultry, and anaerobic digestion are then subtracted. The remaining emissions budget, defined by either a GWP_{100} net-zero target or a split-gas CH_4 target, is used to optimize allowable cattle production through linear programming using Pyomo (Bynum et al., 2021; Hart et al., 2011), subject to land and emissions constraints.

This structure enables systematic exploration of how different land use configurations, abatement measures, and productivity improvements influence Ireland's ability to meet climate goals while maintaining agricultural outputs.

Transitions in agricultural systems are assumed to proceed linearly to 2050, after which activity levels stabilize. In contrast, afforestation and BECCS continue to scale beyond 2050, reflecting realistic deployment timelines for negative emissions technologies.

Three levels of agricultural abatement are included: baseline (no additional measures), MACC-level (full implementation of measures in the Teagasc 2023 MACC (Lanigan et al., 2023)), and "frontier" (a high ambition pathway including grass-clover swards, methane inhibitors, anaerobic digestion, and manure management technologies).

Agricultural productivity can be selected at three tiers, influencing both per-animal and per-hectare outputs. More ambitious scenarios assume higher milk yields and shorter beef finishing times by 2050.

The livestock system is further defined by a user-selected ratio between dairy and suckler beef cows, which determines herd structure and influences emissions and land use.

Finally, optiGOB includes options to expand protein crop areas (e.g., field beans and peas).

Example Output

A brief illustrative example, with a 2050 climate neutrality target year, a 2020 baseline and a split-gas approach, is provided. The CH_4 emissions are reduced by 30% relative to the baseline, while CO_2 and N_2O are balanced under GWP_{100}.

Parameter selection reflects the 'frontier' abatement path, with a strong productivity increase. The dairy unit to beef unit ratio reflects a dairy (10:1) dominated pathway. A higher afforestation rate (16 kha/per year), with a composition of 70:30 conifer to broadleaf split. BECCS and bio-energy (anaerobic digestion and willow) are also included. Wetland restoration is assumed to be 90% of exploited peatland, and 50% of organic soils under grass rewetted. Pig and Poultry output has also been increased by 20%.

Figure 2 shows the emissions and removals by category for the baseline and transition scenario.

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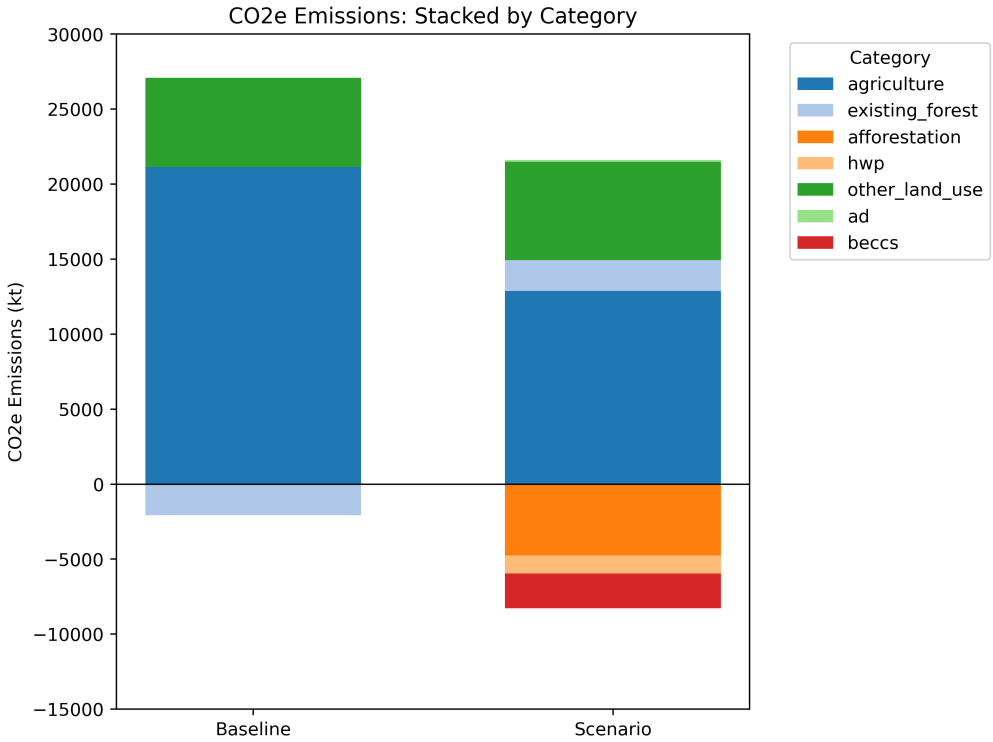

Figure 2: Emissions and Removals from Agriculture, Existing Forest (Managed Forest), Afforestation, Harvested Wood Products (HWP), Other Land Use (Wetlands and Organic Soils), AD (Anaerobic Digestion) and BECCS (CO_2e).

Figure 3 shows total area breakdown by category for the baseline and transition scenario.

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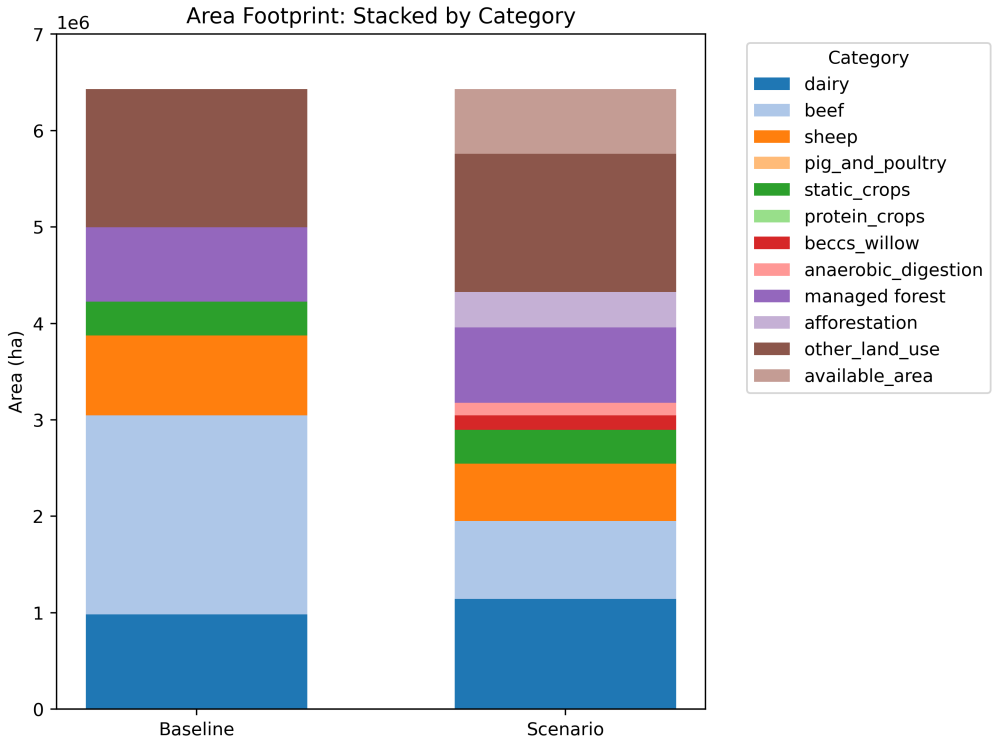

Figure 3: Total area footprint from Dairy, Suckler Beef, Sheep, Pigs & Poultry, Crops, Protein Crops, BECCS (Willow), AD (Anaerobic Digestion), Exiting Forest (Managed Forest), Afforestation, Other Land Use (Wetlands and Organic Soils), and Available Area (Spared area not in use).

Finally, Figure 4 shows the total protein output by category for the baseline and transition scenario. Given the variability in composition of non-protein crops, and the relatively small contribution to the overall protein value, static crops have not been included.

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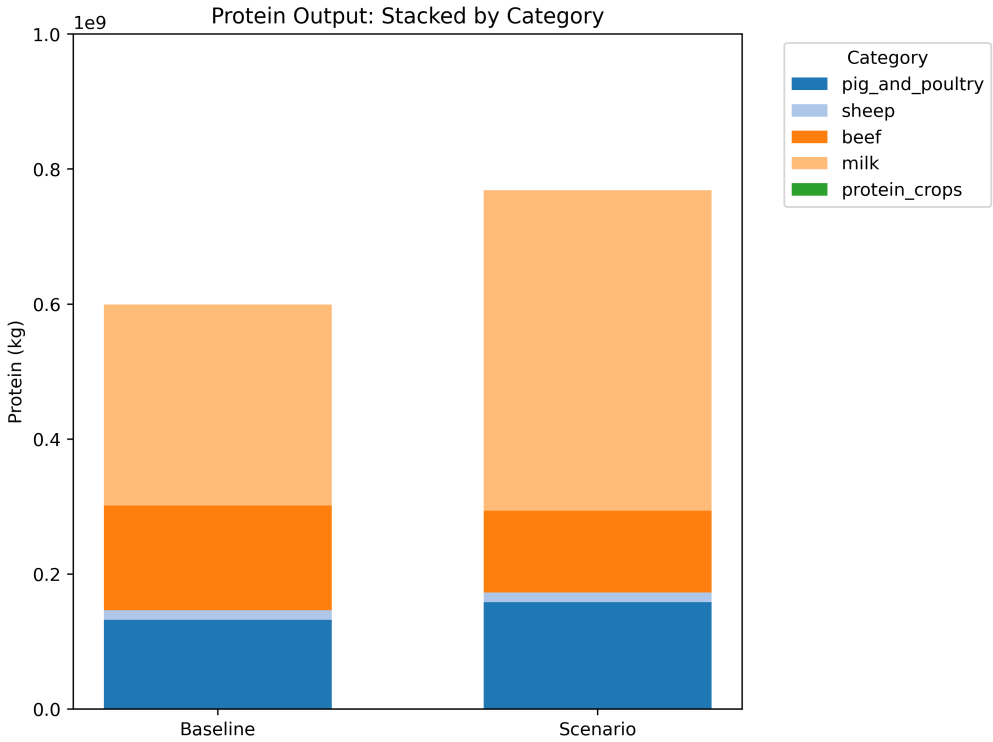

Figure 4: Protein output from Milk, Beef, Pigs & Poultry, Sheep and Protein Crops.

Functionality

The `OptiGob` package provides a flexible decision-support tool to evaluate the environmental and productivity outcomes of transition pathways for Ireland's AFOLU sector. The model integrates modules for livestock, forestry, energy crops, wetlands, and fixed agricultural sectors (e.g., crops, pigs, poultry), enabling users to explore trade-offs and co-benefits across GHG emissions, CDR, land use, and protein production.

Users specify scenario parameters via structured input files (CSV or JSON), including mitigation ambition, productivity levels, afforestation rates, wetland restoration, bioenergy deployment, and livestock restructuring. `OptiGob` calculates resulting emissions, removals, and land requirements under the selected assumptions, using linear optimization to maximize allowable cattle production while meeting emission and land area constraints. The model supports both GWP_{100} net-zero and split-gas targets. However, though the model will optimise livestock population (Dairy and Suckler Beef) to meet emission, livestock ratio and area constraints, this does not ensure that 'net-zero' or split-gas, targets are achieved. The model provides functionality to assess if this has been achieved in a given scenario, but the model does not ensure that the overall combination of land use activities is compliant.

This package is intended to support researchers, policymakers, and students in understanding how different configurations of land use and agricultural change influence Ireland's ability to meet its climate goals. The codebase, available on GitHub (Duffy, 2025), includes worked examples and detailed documentation.

Care should be taken in defining input parameters. A default database is provided, which has several permutations regarding abatement and productivity scenarios. However, the model has a limited number of combinations regarding some inputs, such as rewetting and wetland restoration. A helper method is provided to return valid or supported combinations of input

parameters, particularly for constrained options such as wetland restoration and forestry planting on organic soils. This assists users in constructing valid scenarios and prevents the use of unsupported configurations. Users may also supply their own databases, provided they follow the expected schema.

Limitations

`OptiGob` is designed as a screening and scenario exploration tool with embedded optimisation. rather than a predictive model. It does not guarantee that user-defined combinations of inputs will result in feasible or climate-compliant scenarios; while cattle production is optimised within land and emissions constraints, users must explicitly verify whether their selected climate targets, be they net-zero or split-gas, have been achieved. Some input dimensions, such as wetland restoration and afforestation on organic soils, are limited to specific predefined combinations under the default database, which may restrict flexibility in scenario design. Land use transitions are implemented linearly to 2050, and held constant thereafter, which may not reflect real-world nonlinearities in policy or practice. Emissions and productivity assumptions are based on nationally representative data and mitigation pathways, but do not account for uncertainty or adoption challenges novel technologies. Additionally, the model operates at a national scale and does not include spatial heterogeneity or economic feedbacks. These limitations reflect the model's intended role as a transparent, easy to use, tool for exploring trade-offs and identifying robust configurations under defined assumptions.

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